

Diabetes treatments and risk of amputation, blindness, severe kidney failure, hyperglycaemia, and hypoglycaemia: open cohort study in primary care

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OBJECTIVE To assess the risks of amputation, blindness, severe kidney failure, hyperglycaemia, and hypoglycaemia in patients with type 2 diabetes associated with prescribed diabetes drugs, particularly newer agents including gliptins or glitazones (thiazolidinediones). **DESIGN** Open cohort study in primary care. **SETTING** 1243 practices contributing data to the QResearch database in England. **PARTICIPANTS** 469,688 patients with type 2 diabetes aged 25-84 years between 1 April 2007 and 31 January 2015. **EXPOSURES** Hypoglycaemic agents (glitazones, gliptins, metformin, sulphonylureas, insulin, and other) alone and in combination. **MAIN OUTCOME MEASURES** First recorded diagnoses of amputation, blindness, severe kidney failure, hyperglycaemia, and hypoglycaemia recorded on patients' primary care, mortality, or hospital records. Cox models estimated hazard ratios for diabetes treatments adjusting for potential confounders. **RESULTS** 21,308 (4.5%) and 32,533 (6.9%) patients received prescriptions for glitazones and gliptins during follow-up, respectively. Compared with non-use, glitazones were associated with a decreased risk of blindness (adjusted hazard ratio 0.71, 95% confidence interval 0.57 to 0.89; rate 14.4 per 10,000 person years of exposure) and an increased risk of hypoglycaemia (1.22, 1.10 to 1.37; 65.1); gliptins were associated with a decreased risk of hypoglycaemia (0.86, 0.77 to 0.96; 45.8). Although the numbers of patients prescribed gliptin monotherapy or glitazones monotherapy were relatively low, there were significantly increased risks of severe kidney failure compared with metformin monotherapy (adjusted hazard ratio 2.55, 95% confidence interval 1.13 to 5.74). We found significantly lower risks of hyperglycaemia among patients prescribed dual therapy involving metformin with either gliptins (0.78, 0.62 to 0.97) or glitazones (0.60, 0.45 to 0.80) compared with metformin monotherapy. Patients prescribed triple therapy with metformin, sulphonylureas, and either gliptins (adjusted hazard ratio 5.07, 95% confidence interval 4.28 to 6.00) or glitazones (6.32, 5.35 to 7.45) had significantly higher risks of hypoglycaemia than those prescribed metformin monotherapy, but these risks were similar to those involving dual therapy with metformin and sulphonylureas (6.03, 5.47 to 6.63). Patients prescribed triple therapy with metformin, sulphonylureas, and glitazones had a significantly reduced risk of blindness compared with metformin monotherapy (0.67, 0.48 to 0.94). **CONCLUSIONS** We have found lower risks of hyperglycaemia among patients prescribed dual therapy involving metformin with either gliptins or glitazones compared with metformin alone. Compared with metformin monotherapy, triple therapy with metformin, sulphonylureas, and either gliptins or glitazones was associated with an increased risk of hypoglycaemia, which was similar to the risk for dual therapy with metformin and sulphonylureas. Compared with metformin monotherapy, triple therapy with metformin, sulphonylureas, and glitazones was associated with a reduced risk of blindness. These results, while subject to residual confounding, could have implications for the prescribing of hypoglycaemic drugs.